Five Data Points and a Constant of Nature

Einstein's Photoelectric Equation as a Regression, 1905–1916

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Abstract

In 1905 Einstein wrote down a linear equation relating a measurable voltage to the frequency of light, with a slope fixed in advance by Planck's blackbody constant. In 1916 Millikan measured the slope with five observations and got it right to under one percent. I run the regression, price its standard error, and ask why so few observations sufficed. The answer is that the physics had already fixed the functional form, the number of free parameters, and the units of the slope. A learner starting from the data alone would have needed thousands of observations and would still not have known what it had found. A shorter lesson also emerges: the regression settled the equation and not the physics behind it, and the physics waited another sixty years for a different measurement. A longer version of this note carries the derivations and the full data.

1 The Setup

In December 1900 Max Planck introduced the energy quantum $h\nu$ to fit the blackbody spectrum. He called the step "an act of desperation." Five years later Einstein borrowed the quantum and applied it to a second phenomenon.

Shine light on a metal in a vacuum and electrons come off. Apply a retarding voltage and you can stop them. The voltage that just stops the fastest electron is the *stopping potential* V_0. Einstein proposed that light arrives in discrete quanta of energy $h\nu$, that one quantum is absorbed by one electron, and that the electron escapes if $h\nu$ exceeds the energy W binding it to the metal. Energy conservation then gives

$$V_0 = \frac{h}{e}\,\nu - \frac{W}{e}, \tag{1}$$

where e is the electron's charge. A line. Slope h/e, intercept $-W/e$.

Three features of (1) would interest anyone who estimates equations for a living.

The slope is a universal constant. It does not depend on which metal you use. Every metal must lie on a line of the same gradient, differing only in intercept. Economists will recognise a

*This paper is one of five that read episodes in the history of physics and astronomy as exercises in statistical inference and machine learning. Sargent [2025] sets out the theme: what is now called artificial intelligence names practices that scientists have used for centuries. The other four treat Kepler and Newton, Pauli's exclusion principle, the exoplanet discovery of Mayor and Queloz, and the accelerating universe of Riess, Schmidt and Perlmutter.

cross-equation restriction: one deep parameter is required to appear identically in the reduced form of every experiment.

The slope was known before the experiment. Planck had already fitted h to the blackbody spectrum. Einstein's equation therefore made a prediction with no free parameter to adjust, which is a rarer thing than a good fit.

The intercept is contaminated. Between the metal emitting the electrons and the electrode collecting them sits a contact electromotive force v_c, a voltage difference of purely instrumental origin. It could reach 3 volts, comparable to the stopping potentials themselves, and it drifted with surface age and adsorbed gas. It enters the observed reading additively:

$$V_0^{\mathrm{obs}} = \underbrace{-W/e - v_c}_{\text{intercept}} + \frac{h}{e}\,\nu + \varepsilon. \tag{2}$$

A nuisance parameter that shifts the level and leaves the slope alone. The slope is identified from variation in ν; the intercept is not identified without an independent measurement of v_c.

One more thread runs out of the 1905 paper, and it took twenty years to surface. Einstein's route to the light quantum was thermodynamic. He showed that in one regime the entropy of radiation depends on volume as the entropy of an ideal gas of n independent particles does. He inferred that radiation behaves, for those purposes, as a collection of independent localised quanta. The step from there to treating the classical wave intensity as a probability density for finding a quantum came later, in the 1920s, when Einstein called the wave a *Gespensterfeld*, a ghost field guiding probabilities without carrying energy. Max Born generalised the idea from photons to the Schrödinger wave function in 1926 [Born, 1926] and made it the interpretive core of quantum mechanics. Einstein won the 1921 Nobel Prize for the photoelectric equation. Born won the 1954 Prize for the probabilistic reading that the equation's underlying picture had suggested. Einstein spent the rest of his life opposing it.

The harder case, in the note that follows

The note that follows applies this anatomy to the discovery of cosmic acceleration by Riess, Schmidt and Perlmutter in 1998. General relativity fixes the functional form there, in place of the quantum hypothesis. A combination of the Hubble constant and the supernovae's absolute magnitude plays the part of Millikan's contact voltage, shifting the level of the diagram and leaving its shape alone. The fit again settles a functional form without settling the physics behind it.

The photoelectric case is the easy one, and worth taking first for that reason. Millikan wanted a single number, and removing the nuisance parameter left that number identified. The Friedmann model has two shape parameters. Removing the nuisance parameter there leaves them entangled with each other, and the second note takes up what becomes of identification when it does.

2 Why It Took Eleven Years

Einstein's equation was published in 1905 and confirmed in 1916. The delay was not a failure of statistics. Three groups tried in between and got answers scattered over a factor of two.

Hughes [1913] used three ultraviolet lines spanning 69 nanometres and about a dozen metals, and reported values of h ranging from 3.55 to 5.85×10^{-27} erg·s. Richardson and Compton [1912] measured eight metals and did no better. Every pre-1916 estimate was low by between 12 and 47 percent.

A discrepancy of that size is not sampling error. Three systematic mechanisms produced it. Stray short-wavelength light raised the apparent V_0 at long-wavelength lines and flattened the slope. Contact potentials drifted between sessions, contaminating observations one at a time instead of shifting a common intercept. And the frequency range was too narrow, which inflated the standard error and left the estimate hostage to a single bad line.

Only the third of these is a statistical problem. Estimation theory cannot repair the other two after the fact.

Robert Millikan spent a decade building an apparatus that removed all three. An electromagnetically driven knife shaved a fresh sodium surface inside the vacuum. Filters passed only mercury-arc lines with no short-wavelength companions. A Kelvin double-balance monitored v_c continuously. And the spectral lines spanned 293 nanometres rather than 69.

3 The Regression

Millikan's sodium data, corrected for the contact EMF he measured independently, are in Table 1.

Table 1: Millikan's sodium stopping potentials, corrected for a contact EMF of 2.51 V measured by the Kelvin method. Five of the six values are reconstructions read off the line Millikan fitted to his Figure 5 and rounded to three decimals; only the 2535 Å entry is a directly printed figure.

λ (Å)	ν (10^{14} Hz)	V_0 (V)
5461	5.490	0.454
4339	6.908	1.039
4047	7.409	1.246
3650	8.213	1.578
3126	9.591	2.147
2535	11.827	3.030

Ordinary least squares on the first five points gives

$$\hat{\beta} = \frac{S_{\nu V}}{S_{\nu\nu}} = \frac{3.830}{9.277} = 4.128 \times 10^{-15} \text{ V} \cdot \text{s}, \qquad \hat{\alpha} = -1.813 \text{ V}, \tag{3}$$

where $S_{\nu\nu} = \sum_i (\nu_i - \bar{\nu})^2$ measures how widely the frequencies are spread and $S_{\nu V} = \sum_i (\nu_i - \bar{\nu})(V_{0i} - \bar{V}_0)$ their covariation with the voltages. $S_{\nu\nu}$ carries the whole design argument of Section 4.

The residuals from this fit are all smaller than a quarter of a millivolt, forty times below Millikan's stated reading precision of ± 10 mV. That is not evidence of a very precise experiment. The reconstruction noted in Table 1 produced it. Five of these numbers were read off a fitted line and rounded to three decimals, so the residuals measure rounding. Taken at face value they give a relative standard error of 0.0015%, three hundred times finer than Millikan himself claimed.

The defensible procedure takes σ from the instrument rather than from the residuals. Millikan's balance points were reproducible to about 10 mV, and

$$\mathrm{SE}(\hat{\beta}) = \frac{\sigma}{\sqrt{S_{\nu\nu}}} = \frac{0.010}{\sqrt{9.277 \times 10^{14}}} = 3.3 \times 10^{-17}\ \mathrm{V \cdot s}, \tag{4}$$

a relative uncertainty of 0.8% for the five-line design and 0.5% for the six-line design.

Millikan also determined the slope nine separate times, pairing one spectral line with 5461 Å in nine sessions with a freshly shaved surface each time. Those nine values are genuine replications. Their mean is 4.131×10^{-15}, their standard deviation 0.087, and the standard error of the mean 0.029, or 0.71%.

Two routes, one from replication and one from the information matrix, agree to within a tenth of a percent. That agreement, and not the spurious 0.0015%, is what the uncertainty on h rests on.

The cross-metal restriction

Because h/e is universal, every metal must give the same slope. Millikan also ran lithium. Fitting five lithium lines gives $\hat{\beta}_{\mathrm{Li}} = 4.127 \times 10^{-15}$, differing from the sodium slope by 0.04%. Pooling the two metals with a common slope and separate intercepts gives 4.127×10^{-15} with a standard error of 0.49%, since pooling enlarges $S_{\nu\nu}$ in the same way that adding the sixth sodium line does. The Wald statistic for equality of the two slopes is 0.04.

The restriction is not rejected, and it would be a mistake to call that a confirmation. The test has little power. With a standard error of 0.8% on each slope it would fail to reject differences of up to about 2%, so it excludes only gross violations. More seriously, four of the five lithium entries in Millikan's table are the sodium entries shifted by a constant 0.540 V, and a constant shift leaves the slope unchanged by construction. The near-zero test statistic reflects the arithmetic as much as the physics. Millikan established that the two metals' curves were parallel at his stated precision, which is how he put it.

4 Design Beats Sample Size

Under Gaussian errors, maximising the likelihood and minimising the sum of squared residuals are the same operation, so the least-squares estimates in (3) are also maximum likelihood estimates. The Fisher information for (α, β) is $\sigma^{-2} X^{\top} X$, and inverting it gives the Cramér–Rao bound on the variance of any unbiased estimator of the slope,

$$\mathrm{Var}(\tilde{\beta}) \ \geq\ \frac{\sigma^2}{S_{\nu\nu}}, \qquad S_{\nu\nu} = \sum_i (\nu_i - \bar{\nu})^2, \tag{5}$$

which least squares attains exactly. Nothing can do better.

Equation (5) is a statement about the *design* and not about the readings. $S_{\nu\nu}$ depends only on which spectral lines the experimenter chose. Spreading the frequencies apart shrinks the bound. Collecting more readings at the same frequencies does much less.

Table 2: Effect of the frequency range on the standard error of h/e, holding the instrumental $\sigma = 10$ mV fixed. The narrow window is the closest analogue Millikan's line set permits to the bands available to Hughes and to Richardson & Compton.

Design	n	Span (nm)	$S_{\nu\nu}$ (10^{28} Hz^2)	Rel. SE
Narrow (4047–3126 Å)	3	92	2.44	1.55%
Millikan 5-point	5	234	9.28	0.79%
Millikan 6-point	6	293	24.72	0.49%

Going from three lines to six cuts the standard error by a factor of three, and almost all of the gain comes from the wider span rather than from the extra observations. The point generalises past photoelectrics. Where a parameter is identified by variation in a regressor, the design of that variation dominates the sample size.

The sixth line illustrates the cost of the same strategy. Adding the 2535 Å point, the only directly quoted number in the table, moves the slope to 4.070×10^{-15}, 1.4% down. It sits 0.04 V below the five-point line, and its extreme frequency gives it the highest leverage. Millikan attributed the deficit to scattered ultraviolet light and dropped the line from his primary determination. The observation that contributes most to $S_{\nu\nu}$ is also the one most exposed to systematic error.

5 The Prediction

Converting the slope to Planck's constant needs the electron charge. With the value Millikan had measured himself, $e = 1.592 \times 10^{-19}$ C, the five-point slope gives $h = 6.573 \times 10^{-34}$ J·s, low by 0.81% against the modern defined value. With the modern e it gives 6.616×10^{-34}, low by 0.17%. Millikan's error came from his oil-drop charge, which was too small by 0.64%, and not from the photoelectric slope his 1916 apparatus was built to measure.

The sharper test does not involve the modern value at all. Planck had fitted $h = 6.55 \times 10^{-27}$ erg·s to the blackbody spectrum in 1901. With Millikan's own e, that predicts a slope

$$\left(\frac{h}{e}\right)_{\text{predicted}} = \frac{6.55 \times 10^{-34}}{1.592 \times 10^{-19}} = 4.114 \times 10^{-15} \text{ V} \cdot \text{s}. \tag{6}$$

Millikan measured 4.128×10^{-15}. The two agree to 0.35%, inside the 0.8% standard error.

No photoelectric measurement entered the prediction. Planck fitted h to a different phenomenon fifteen years earlier. Einstein derived the slope in 1905. Millikan measured it in 1916. The regression had one free parameter, the intercept, and it was not the one being tested.

Both 1916-era constants were too small, and the errors partly cancel in the ratio. Planck's h was low by 1.15% and Millikan's e by 0.64%, so the predicted slope was low by 0.52% and the measured slope by 0.17%. Against modern constants the comparison is 4.128 against 4.136.

6 The Equation Is Still in Service

Equation (1) did not become a historical curiosity. Laboratories with no interest in Planck's constant read it backwards every day.

In X-ray photoelectron spectroscopy the incident photon energy is known and the measured electron energy identifies the binding energy of the core electron the photon knocked loose:

$$E_B = h\nu - eV_0 - \phi_{\text{spec}}, \tag{7}$$

where ϕ_{spec} is an instrument calibration constant. That is (1) rearranged, with a different unknown. Because every element has its own set of core-level binding energies, the technique reads the elemental composition of a surface directly. Kai Siegbahn received the 1981 Nobel Prize for it [Siegbahn et al., 1967], and it is now standard in semiconductor fabrication, catalysis and materials science. Ultraviolet photoelectron spectroscopy and Kelvin probes apply the same equation to measure work functions of semiconductor films and photovoltaic layers to a few meV.

Einstein's equation survives in quantum electrodynamics unchanged, as the energy balance for the fastest photoelectron: the electron that starts at the Fermi level and reaches the detector without inelastic loss. The balance is exact. The measured spectrum is not. A real metal broadens the emission edge thermally, the work function belongs to a crystal face and its adsorbate layer rather than to an element, and final-state relaxation shifts core levels by amounts that matter at spectroscopic precision. Millikan's residual scatter came from the surface physics. The energy balance was never the problem.

7 The Man Who Confirmed It Did Not Believe It

Millikan's 1916 paper confirmed Einstein's equation to better than one percent. In the same paper he wrote:

> "Despite then the apparently complete success of the Einstein equation, the physical theory of which it was designed to be the symbolic expression is found so untenable that Einstein himself, I believe, no longer holds to it." [Millikan, 1916, p. 388]

He had built the apparatus hoping to *refute* the equation. The light-quantum hypothesis struck him as physically impossible, since a stream of localised lumps could not diffract or interfere, and diffraction and interference were not in doubt. He got the opposite of what he expected and published it anyway, with the interpretation disowned.

The stance is more defensible than it sounds, and it will be familiar to anyone who has estimated a reduced form. Millikan confirmed a predictive relationship without endorsing the structural model that generated it. He was right to separate the two, and Section 9 shows that he was right on the merits as well: the regression really does not establish the mechanism. Where he erred was in thinking the equation's success left the hypothesis untouched.

Compton scattering [Compton, 1923] supplied independent evidence for the particulate character of light in 1923. Millikan came to describe his own result differently:

> "This work resulted, contrary to my own expectation, in the first direct experimental proof . . . of the complete validity of the Einstein equation."

The sentence belongs to his later retrospective writing rather than to the Nobel lecture he delivered in 1924. Holton [1999] and Pais [1982] trace how his views moved.

8 Einstein as Machine Learner

Recast the episode in the vocabulary of statistical learning. The translation is not a conceit. Sargent [2025] argues that the techniques now gathered under artificial intelligence have long histories inside the sciences that used them, and that reading old episodes this way shows what the methods can and cannot do.

The *hypothesis class* is the set of functions the estimator is allowed to consider. Einstein's is

$$\mathcal{H} = \{\nu \mapsto \alpha + \beta\nu\}, \tag{8}$$

with β shared across metals. Two free parameters per metal, one shared slope, pseudo-dimension two. The restriction was not a modelling convenience. It followed deductively from $E = h\nu$ and energy conservation. The quantum hypothesis *is* the inductive bias.

The *feature selection* was Lenard's. In 1902 he established that the stopping potential does not depend on light intensity and does depend on frequency. Classical wave theory predicted the opposite. Lenard's experiment removed intensity from the feature set before any line was fitted.

The *training* is least squares, and for this class it has a closed form. No gradient descent, no learning rate, no early stopping.

Now compare the alternative. A three-layer network with sixteen ReLU units per layer has

$$p = (1\cdot 16 + 16) + (16\cdot 16 + 16) + (16\cdot 16 + 16) + (16\cdot 1 + 1) = 593$$

free parameters and a pseudo-dimension near 5,500. The pseudo-dimension is the capacity measure appropriate to real-valued function classes; for a linear model it equals the number of free parameters, which is why Einstein's class scores two. The practitioner's rule of thumb, $n \gtrsim 10p$, calls for about six thousand training examples. Millikan had five.

Table 3: What was available when, against what each approach needs. The network's requirement uses $n \gtrsim 10p$; the formal uniform-convergence bound is four orders of magnitude larger and vacuous here.

Epoch	n	Einstein's model	Neural network
Lenard 1902	≈ 0	Sufficient†	Cannot train
Hughes 1912	~ 30	Sufficient	Short by $\times 200$
Millikan 1916	5	Sufficient	Short by $\times 1200$
Modern era	~ 5000	Sufficient	Short by $\times 1.2$

† Theory fixes the functional form; two parameters remain to be estimated from data.

The last row is the interesting one. Given enough clean observations a network would eventually trace out the line. It would still not identify the fitted gradient as the ratio h/e of two independently measured constants of nature. It would still not impose the cross-metal restriction from first principles, seeing universality as a coincidence to be discovered rather than a constraint to be applied. And it would not decompose the intercept into a work function and an apparatus nuisance, because those two are not separately identified without the external Kelvin measurement.

Identification failures of this kind are invisible to a fitting procedure. The regression runs, reports a number, and says nothing about what the number means.

Table 4 sets the two vocabularies side by side.

Table 4: The photoelectric episode in two vocabularies.

Step	Scientific name	Machine-learning name
Propose $E = h\nu$	Theoretical hypothesis	Model architecture
Derive $V_0 = (h/e)\nu - W/e$	Deduction	Hypothesis class
Use ν, not intensity	Feature selection	Feature engineering
Measure (ν_i, V_{0i})	Experiment	Training data
Minimise $\sum \hat{\varepsilon}_i^2$	Least squares	Squared-error loss
Slope matches Planck's h/e	Prediction confirmed	Out-of-sample test
Na and Li share a slope	Universality	Parameter tying

Five points sufficed because the physics had already supplied the functional form, the parameter count, and the units of the slope. Theory does not substitute for data. It compresses how much data you need.

Where the compression runs out

The compression is not uniform across the parameters, and the intercept shows where it fails. Millikan's fitted intercept implies a sodium work function of $-\hat{\alpha} = 1.81$ eV. The modern value for clean bulk sodium is 2.36 eV.

Part of the gap is estimation error. The intercept extrapolates to $\nu = 0$, far outside the observed range, and its standard error is about 0.08 V at $\sigma = 10$ mV, an order of magnitude worse in relative terms than the slope's. The rest is physics. Millikan's freshly shaved surfaces sat in residual gas, and adsorbed gas shifts a metal's work function by several tenths of a volt. The work function also varies by some 0.3 eV across the crystal faces of a single metal. Sodium has no one work function for the intercept to be compared against.

The two coefficients are identified on different terms. The slope is a universal constant, identified by variation within a run, and immune to any disturbance common to all the lines. The intercept is a surface-specific quantity, identified by extrapolation, and contaminated by exactly the conditions the apparatus could not control. Einstein's equation is far better identified in its slope than in its intercept. The slope carries the physics.

9 What the Regression Could Not Settle

The fit is excellent and the prediction was confirmed. Neither establishes that light is made of photons.

Lamb and Scully [1969] showed that equation (1) follows from a *semiclassical* model in which the electromagnetic field stays a classical wave and only the atom is quantised. The equation is a statement of energy conservation together with quantised *atomic* energy levels. It does not require the radiation field itself to come in lumps.

So Millikan's regression selected a functional form and ruled out a rival form. It could not choose between two physical stories that imply the same form. A model-selection criterion scoring in-sample fit would have reported overwhelming evidence and would have been answering a different question from the one everyone thought was being asked.

Settling it took a different observable and sixty more years. The decisive evidence is photon *antibunching*: photons from a single quantum emitter arrive *less* bunched than Poisson statistics allow. No classical stochastic field can produce sub-Poissonian counts, however it is modulated. Kimble et al. [1977] observed the effect in the resonance fluorescence of sodium atoms.

The experiment was designed to test field quantisation, and the observable it measured, photon-number variance, follows from Glauber's quantum theory of optical coherence. Without that theory nobody would have thought to measure photon-number correlations at all.

The photoelectric episode is usually told as a confirmation. A division of labour describes it better.

10 What This Says About AI in the Sciences

Deep networks are the right tool for problems in which no parsimonious parametric family is known. Image recognition has no closed form. Neither does protein structure prediction, and AlphaFold's achievement was to learn a map that six decades of physical modelling had failed to write down. Neither does language. In each case the hypothesis class is genuinely unknown, the input is high-dimensional, and data are abundant. Learning the function is the only available move.

The photoelectric problem sits at the opposite corner of the design space: one input, one output, five observations, and a functional form already derived from a physical principle. Nothing about the success of learned models on the first class of problem bears on the second.

Two claims often made for AI in the sciences are worth separating against this example.

The first is that a sufficiently capable learner can rediscover physical laws from data. In a limited sense this is true and has been demonstrated. Symbolic regression systems such as AI Feynman [Udrescu and Tegmark, 2020] recover the functional forms of textbook relations from tabulated data. But they are given the units and the physical constants as inputs, and it is the units that let a fitted coefficient be matched to h/e rather than reported as an unexplained number. That hardcoding is prior physical knowledge. The (ν, V_0) data do not contain it.

The second is that scaling data collection substitutes for theory. The Cramér–Rao bound in (5) says otherwise, and says it precisely. The standard error falls with the *spread* of the regressor, not with the count of observations at fixed design. Millikan could have taken ten thousand readings inside the 92-nanometre window and done worse than five readings across 293 nanometres. Deciding where to look is a modelling decision, and the model that tells you where to look is the thing being tested.

The sharper limit is the one in Section 9. Even a perfect fit of a correctly specified equation to unlimited data would not have settled whether light is quantised, because a rival physical theory implies the same equation. Getting that question answered required somebody to prove that the two theories differ in photon-number variance, and then somebody to measure photon-number variance. No quantity of (ν, V_0) pairs contains the answer. A system that optimises fit on the data it is given cannot discover that it is measuring the wrong thing.

11 Summary

Einstein's photoelectric equation is a two-parameter regression whose slope was fixed in advance by a constant fitted to an unrelated phenomenon. Five observations measured it to under one percent. The regression itself is trivial; obtaining clean data took a decade of instrument building, and widening the range of the regressor mattered more than adding observations. The final estimates are

$$\hat{\beta} = 4.128 \times 10^{-15}\ \mathrm{V \cdot s} \pm 0.8\%, \qquad h = 6.62 \times 10^{-34}\ \mathrm{J \cdot s}, \qquad W_{\mathrm{Na}}/e = 1.81\ \mathrm{eV}.$$

Two results point in opposite directions. A correctly specified model turned a problem needing thousands of observations into one needing five. The same model, fitted to those five points as precisely as anyone could wish, left the physical question open for another sixty years, until a theory arrived that named a different thing to measure.

Millikan's tabulated stopping potentials, apart from the 2535 Å entry, are reconstructions rather than independently recovered readings, and the standard errors quoted here come from his instrumental precision and his replicated sessions rather than from the residuals of the fitted line. Planck's 1901 value for h and the sourcing of the Millikan quotations come from secondary sources and should be checked against the originals.

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